

# **DB Schema**

## **1. users**

- user\_id (string, PK)
- account\_age\_days (int)
- email\_verified (bool/int)
- shared\_device\_flag (bool/int)

## **2. sellers**

- seller\_id (string, PK)
- seller\_age\_days (int)
- seller\_rating (float)
- seller\_return\_rate (float)
- seller\_customer\_frequency (float)
- seller\_type (int, FK(seller\_type\_id) from seller\_types table)
- category\_specialization (string)
- rating (float)

## **3. products**

- product\_id (string, PK)
- category (int, FK(product\_category\_id) from product\_categories table)
- price (double)
- is\_non\_returnable (int/bool)
- review\_count (int)
- sku (int)
- category\_base\_defect\_rate (double)
- return\_window\_days (int)
- base\_defect\_rate (float)

## **4. orders**

- order\_id (string, PK)
- user\_id (string, FK from users table)
- seller\_id (string, FK from sellers table)
- product\_id (string, FK from products table)
- order\_value (double)
- is\_prepaid (int/bool)
- discount\_pct (float)
- order\_date (date, YYYY-MM-DD)
- delivery\_date (date, YYYY-MM-DD)
- is\_returned (int/bool)

## **5. returns**

- return\_id (string, PK)
- order\_id (string, FK from orders table)
- reason\_category (int, FK(reason\_category\_id) from reason\_categories table)
- return\_type (int, FK(return\_type\_id) from return\_type table)
- image\_uploaded (int/bool)

- return\_request\_date (date)
- within\_return\_window (int, bool)
- days\_left\_to\_return (int)
- total\_orders\_at\_time (int)
- total\_returns\_at\_time (int)
- fraud\_type (int, FK(fraud\_id) from fraud\_type table)
- label (int, FK(label\_id) from label\_type table)

#### **6. fraud\_type**

- fraud\_id (int, PK)
- fraud\_name (string)

#### **7. label\_type**

- label\_id (int, PK)
- label\_name (string)

#### **8. return\_type**

- return\_type\_id (int, PK)
- return\_type\_name (string)

#### **9. product\_categories**

- product\_category\_id (int, PK)
- product\_category\_name (string)

#### **10. seller\_types**

- seller\_type\_id (int, PK)
- seller\_type\_name (string)

#### **11. reason\_categories**

- reason\_category\_id (int, PK)
- reason\_category\_name (string)

#### **12. adversarial\_probe\_set**

- case\_id (string, PK)
- bucket (int, FK(bucket\_type\_id) from bucket\_type table)
- category (int, FK(product\_category\_id) from product\_categories table)
- reason\_category (int, FK(reason\_category\_id) from reason\_categories table)
- account\_age\_days (int)
- total\_orders\_at\_time (int)
- total\_returns\_at\_time (int)
- return\_to\_order\_ratio (float)
- days\_since\_last\_return (int)
- returns\_last\_30d (int)
- returns\_last\_90d (int)
- order\_value (int)
- discount\_pct (int)
- order\_value\_ratio (float)
- seller\_return\_rate (float)
- seller\_age\_days (int)
- shared\_device\_flag (int/bool)

- email\_verified (int/bool)
- phone\_verified (int/bool)
- image\_uploaded (int/bool)
- within\_return\_window (int/bool)
- days\_left\_to\_return (int)
- is\_non\_returnable (int/bool)
- narrative (string)
- expected\_verdict (int, FK(verdict\_type\_id) from verdict\_type table)
- expected\_reasoning\_tag (string)
- notes (string)
- image\_path (string)
- customer\_custom\_text (string)
- assigned\_user\_id (string)
- assigned\_seller\_id (string)
- assigned\_product\_id (string)
- assigned\_order\_id (string)
- assigned\_return\_id (string)

### 13. bucket\_type

- bucket\_type\_id (int, PK)
- bucket\_type\_name (string)

### 14. verdict\_type

- verdict\_type\_id (int, PK)
- verdict\_type\_name (string)

## PlantUML code for generating ER diagram

```
@startuml
'=====
' GRID - Fraud Returns Database
'=====

hide circle
hide methods
skinparam linetype ortho
skinparam shadowing false
skinparam dpi 220
skinparam defaultFontName Arial

entity "users" as users {
    * user_id : varchar <<PK>>
```

```

--
account_age_days : int
email_verified : boolean
shared_device_flag : boolean
registration_date : text
phone_verified : text
}

entity "seller_types" as seller_types {
  * seller_type_id : int <<PK>>
  --
  seller_type_name : varchar
}

entity "product_categories" as product_categories {
  * product_category_id : int <<PK>>
  --
  product_category_name : varchar
}

entity "sellers" as sellers {
  * seller_id : varchar <<PK>>
  --
  seller_age_days : int
  seller_rating : float
  seller_return_rate : float
  seller_customer_frequency : float
  seller_type_id : int <<FK>>
  registration_date : text
  category_specialization_id : int <<FK>>
  rating : float
}

entity "products" as products {
  * product_id : varchar <<PK>>
  --
  product_category_id : int <<FK>>
  price : decimal
  is_non_returnable : boolean
  review_count : int
  sku : int
  category_base_defect_rate : float
  return_window_days : int
  base_defect_rate : float
}

```

```
}
```

```
entity "orders" as orders {  
    * order_id : varchar <<PK>>  
    --  
    user_id : varchar <<FK>>  
    seller_id : varchar <<FK>>  
    product_id : varchar <<FK>>  
    order_value : decimal  
    is_prepaid : boolean  
    discount_pct : float  
    order_date : date  
    delivery_date : date  
    is_returned : boolean  
}
```

```
entity "reason_categories" as reason_categories {  
    * reason_category_id : int <<PK>>  
    --  
    reason_category_name : varchar  
}
```

```
entity "return_type" as return_type {  
    * return_type_id : int <<PK>>  
    --  
    return_type_name : varchar  
}
```

```
entity "fraud_type" as fraud_type {  
    * fraud_id : int <<PK>>  
    --  
    fraud_name : varchar  
}
```

```
entity "label_type" as label_type {  
    * label_id : int <<PK>>  
    --  
    label_name : varchar  
}
```

```
entity "returns" as returns {  
    * return_id : varchar <<PK>>  
    --  
    order_id : varchar <<FK>>
```

```

reason_category_id : int <<FK>>
return_type_id : int <<FK>>
image_uploaded : boolean
return_request_date : date
within_return_window : boolean
days_left_to_return : int
total_orders_at_time : int
total_returns_at_time : int
fraud_id : int <<FK>>
label_id : int <<FK>>
}

entity "bucket_type" as bucket_type {
  * bucket_type_id : int <<PK>>
  --
  bucket_type_name : varchar
}

entity "verdict_type" as verdict_type {
  * verdict_type_id : int <<PK>>
  --
  verdict_type_name : varchar
}

entity "adversarial_probe_set" as probe {
  * case_id : varchar <<PK>>
  --
  bucket_type_id : int <<FK>>
  product_category_id : int <<FK>>
  reason_category_id : int <<FK>>

  account_age_days : int
  total_orders_at_time : int
  total_returns_at_time : int
  return_to_order_ratio : float
  days_since_last_return : int
  returns_last_30d : int
  returns_last_90d : int

  order_value : decimal
  discount_pct : float
  order_value_ratio : float

  seller_return_rate : float

```

seller\_age\_days : int

shared\_device\_flag : boolean

email\_verified : boolean

phone\_verified : boolean

image\_uploaded : boolean

within\_return\_window : boolean

days\_left\_to\_return : int

is\_non\_returnable : boolean

narrative : text

expected\_reasoning\_tag : text

notes : text

image\_path : text

customer\_custom\_text : text

assigned\_user\_id : varchar <<FK>>

assigned\_seller\_id : varchar <<FK>>

assigned\_product\_id : varchar <<FK>>

assigned\_order\_id : varchar <<FK>>

assigned\_return\_id : varchar <<FK>>

verdict\_type\_id : int <<FK>>

}

'=====

' Relationships

'=====

seller\_types ||--o{ sellers

product\_categories ||--o{ sellers

product\_categories ||--o{ products

product\_categories ||--o{ probe

users ||--o{ orders

sellers ||--o{ orders

products ||--o{ orders

orders ||--o{ returns

reason\_categories ||--o{ returns

return\_type ||--o{ returns

fraud\_type ||--o{ returns  
label\_type ||--o{ returns

bucket\_type ||--o{ probe  
reason\_categories ||--o{ probe  
verdict\_type ||--o{ probe

users ||--o{ probe : assigned\_user  
sellers ||--o{ probe : assigned\_seller  
products ||--o{ probe : assigned\_product  
orders ||--o{ probe : assigned\_order  
returns ||--o{ probe : assigned\_return

@enduml